 import pandas as pd – imports Pandas library.

```
import pandas as pd
```

```
df = pd.read_csv("Students.csv")
```

 head() – displays the first records.

```
print(df.head())
```

```
   Roll No    Name Department  Marks  Attendance
0        1  khatri     BSc IT     78          85
1        2   dalla     BSc CS     65          90
2        3   munna     BSc IT     88          92
3        4    golu     BSc CS     72          80
4        5   lallu     BSc IT     55          70
```

 tail() – displays the last records.

```
print(df.tail())
```

```
   Roll No    Name Department  Marks  Attendance
0        1  khatri     BSc IT     78          85
1        2   dalla     BSc CS     65          90
2        3   munna     BSc IT     88          92
3        4    golu     BSc CS     72          80
4        5   lallu     BSc IT     55          70
```

shape – gives the number of rows and columns.

```
print(df.shape)
```

```
(5, 5)
```

```
print(df.columns)
```

```
Index(['Roll No', 'Name', 'Department', 'Marks', 'Attendance'], dtype='object
```

dtypes – displays data types of columns.

```
print(df.dtypes)
```

```
Roll No        int64
Name          object
Department    object
Marks          int64
```

```
Attendance     int64
dtype: object
```

 describe() – displays basic statistical information.

```
print(df.describe)
```

```
<bound method NDFrame.describe of    Roll No    Name Department  Marks  Atten
0        1  khatri     BSc IT     78          85
1        2   dalla     BSc CS     65          90
2        3   munna     BSc IT     88          92
3        4    golu     BSc CS     72          80
4        5   lallu     BSc IT     55          70>
```

info() – displays complete information about the dataset.

```
print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 5 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Roll No     5 non-null      int64
 1   Name        5 non-null      object
 2   Department  5 non-null      object
 3   Marks       5 non-null      int64
 4   Attendance  5 non-null      int64
dtypes: int64(3), object(2)
memory usage: 332.0+ bytes
None
```

6c. Filter Dataset Use df["Column"] to access a particular column.

```
col = df["Marks"]
print(col)
```

```
0    78
1    65
2    88
3    72
4    55
Name: Marks, dtype: int64
```

 Use conditions such as >, <, ==, >=, and <= to filter records.

```
filtered = df[
    (df["Marks"] > 70) & (df["Department"] == "BSc IT")
]
print(filtered)
```

```
   Roll No    Name Department  Marks  Attendance
0        1  khatri     BSc IT     78          85
2        3   munna     BSc IT     88          92
```

6d. Sort Dataset sort_values() is used to arrange records according to a column.

```
sorted_df = df.sort_values(
    by="Marks", ascending=True
)  # Marks ke according sort (ascending)
print(sorted_df)
```

```
   Roll No    Name Department  Marks  Attendance
4        5   lallu     BSc IT     55          70
1        2   dalla     BSc CS     65          90
3        4    golu     BSc CS     72          80
0        1  khatri     BSc IT     78          85
2        3   munna     BSc IT     88          92
```

```
print("Maximum Marks ",df["Marks"].max())
print("Minimum Marks ",df["Marks"].min())
```

```
Maximum Marks  88
Minimum Marks  55
```